

AI-Command-Center.md

AI Command Center

ArenaMind AI UI/UX Specification

Screen ID: ACC-001

Version: 1.0

Status: Draft

Priority: ★★★★★

Overview

The **AI Command Center** is the operational heart of ArenaMind AI.

It serves as the primary workspace for stadium operators, security teams, tournament organizers, emergency responders, transportation coordinators, and executive decision-makers.

Unlike conventional dashboards, the AI Command Center is an **AI-native operational workspace** where Generative AI continuously observes, analyzes, predicts, explains, and recommends actions.

The interface should feel like the **Mission Control Center of the FIFA World Cup 2026**.

Design Goals

The screen must enable operators to answer four critical questions instantly:

- **What is happening now?**
- **What will happen next?**
- **What requires immediate attention?**
- **What actions should be taken?**

The Command Center should prioritize clarity, confidence, and speed.

Primary Users

- Stadium Operations Manager
- Security Control Room
- Emergency Response Teams
- Transportation Coordinator
- Volunteer Coordinator
- FIFA Match Operations
- Executive Leadership

Layout Overview

Floating Navigation + AI Status + Global Search		
Left Panel	Main Workspace	Right Panel
Navigation		AI Copilot
Quick Tools	Digital Twin + Live Map	Recommendations
Alerts	Crowd Intelligence	Incidents
Shortcuts	Operational Widgets	Notifications
Bottom Command Dock (Natural Language AI + Voice + Actions)		

Information Architecture

```
AI Command Center
|
├─ Global Navigation
├─ AI Operational Summary
├─ Stadium Digital Twin
├─ Live Crowd Intelligence
```

- | Incident Timeline
- | Transportation Status
- | Emergency Monitor
- | Volunteer Operations
- | Sustainability Metrics
- | AI Copilot
- | Notification Center
- | Command Dock

Screen Sections

1. Global Navigation Bar

Persistent floating navigation.

Left

- ArenaMind Logo
- Stadium Selector
- Match Selector

Center

Global AI Search

Example:

```
Ask ArenaMind...  
  
"Predict congestion near Gate B"  
  
"Summarize incidents"  
  
"Show medical alerts"
```

Right

- User Avatar
- Notifications

- Settings
 - Language Selector
 - AI Status
-

AI Status Indicator

Displays

 Gemini Online

Fallback Ready

Groq Available

Latency

420 ms

If Gemini quota is exceeded:

 Switching to Groq...

No interruption.

Transition should occur seamlessly.

2. Left Navigation Rail

Glassmorphism sidebar.

Icons only when collapsed.

Expanded view includes labels.

Modules

- Dashboard
- Digital Twin

- Crowd
 - Transport
 - Volunteers
 - Emergency
 - Sustainability
 - Analytics
 - Reports
 - Settings
-

Quick Actions

Pinned shortcuts.

Generate Report

Broadcast Alert

Simulate Incident

Open Camera

Assign Volunteers

View Heatmap

3. AI Operational Summary

Displayed immediately after login.

Example

Good Evening, Operations Team.

Current Stadium Status

Operational

96%

No critical incidents.

Two congestion risks detected.

Rain expected in 40 minutes.

Transport operating normally.

Generated automatically by AI.

Updates continuously.

4. Digital Twin Workspace

Primary visual focus.

Occupies approximately 45% of the viewport.

Interactive 3D stadium supporting:

- Rotate
- Zoom
- Select zones
- Layer toggles
- AI overlays

Layers

- Crowd Density
- Security
- Medical
- Volunteers
- Weather
- Parking

- Accessibility
- Energy
- Transport

Users can enable multiple layers simultaneously.

Zone Interaction

Selecting a stadium zone opens a contextual panel.

Example

Gate B

Current Occupancy

81%

Prediction

High congestion in 11 minutes.

Confidence

94%

Recommendation

Redirect spectators to Gate D.

Assign 4 volunteers.

Display multilingual guidance.

5. Operational KPI Strip

Horizontal metrics.

Metric	Example
Attendance	67,842

Metric	Example
Occupancy	82%
Crowd Index	Moderate
Incidents	2 Active
Volunteers	1,284
Energy Usage	74%
Weather Risk	Low
AI Confidence	97%

Animated number transitions.

6. Live Incident Timeline

Chronological feed.

Each card includes:

- Timestamp
- Location
- Category
- Severity
- AI Summary
- Assigned Team
- Resolution Status

Example

18:42

Gate C

Crowd Congestion

Medium

AI predicts queue growth within 8 minutes.

Assigned

Operations Team

7. AI Recommendation Panel

Right-side floating panel.

Displays prioritized recommendations.

Structure

Priority

Recommendation

Reasoning

Confidence

Expected Impact

Execute

Example

High Priority

Open Gate D

Reason

Bus arrival detected.

Confidence

96%

Impact

Reduce queue by 31%.

[Execute]

8. AI Copilot

Persistent assistant.

Supports

- Text
- Voice
- Image
- PDF
- Camera metadata

Example prompts

Summarize stadium status.

Show all security incidents.

Generate executive briefing.

Predict crowd movement.

Simulate emergency evacuation.

Streaming responses.

Supports follow-up conversations.

9. Notification Center

Grouped by severity.

Critical

Warning

Information

Success

Supports:

- Pin
- Archive
- Filter
- Acknowledge

10. Bottom Command Dock

Primary interaction model.

Ask ArenaMind...



Supports

- Natural Language
- Voice Commands
- File Upload
- Camera Snapshot
- Operational Commands

AI Workflow Example

Operator asks:

Predict congestion near Gate A.

ArenaMind

1. Queries Gemini.
 2. Retrieves live crowd data.
 3. Applies RAG with stadium policies.
 4. Generates prediction.
 5. If Gemini unavailable, automatically routes to Groq.
 6. Streams response.
 7. Highlights affected zone.
 8. Displays confidence score.
 9. Suggests actions.
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AI Confidence Component

Every AI response includes:

Confidence

97%

Based On

CCTV Metadata

Ticket Scans

Weather

Historical Patterns

Transport Data

Transparency is mandatory.

AI Explanation Panel

Every recommendation answers:

- Why?
 - What evidence?
 - What happens if ignored?
 - What alternatives exist?
-

Command Palette

Shortcut

⌘ K

Ctrl + K

Examples

Go to Digital Twin

Generate Report

Search Incident

Open Gate B

Switch Language

View Cameras

Component Inventory

- Floating Navigation
- Global Search
- Digital Twin Viewer
- AI Summary Card
- KPI Cards

- Incident Cards
 - AI Recommendation Cards
 - Notification Panel
 - Command Dock
 - AI Chat
 - Glass Sidebar
 - Heatmap Widget
 - Weather Widget
-

Motion Specifications

Navigation

Glass blur

↓

Slide

↓

Fade

Digital Twin

Continuous rotation

↓

Focus animation

↓

Zone highlight

AI Responses

Thinking animation

↓

Streaming response

↓

Confidence reveal

Notifications

Slide

↓

Glow

↓

Expand

Accessibility

Supports

- Keyboard navigation
 - Screen readers
 - Voice control
 - High contrast mode
 - Reduced motion
 - Focus indicators
 - WCAG 2.2 AA
-

Responsive Design

Desktop

Three-column mission control layout.

Tablet

Digital Twin above AI panel.

Mobile

Stacked cards.

Bottom AI dock remains persistent.

Empty States

If no incidents exist:

● Stadium operating normally.

No active incidents.

AI continues monitoring live systems.

Loading States

Never use generic spinners.

Display:

- Stadium wireframe
 - Neural pulse
 - AI reasoning timeline
 - Skeleton cards
-

Error Handling

If Gemini returns a rate-limit response:

Gemini temporarily unavailable.

Switching to Groq...

Operational context preserved.

No user interaction required.

Performance Targets

Metric	Target
Initial Load	<3 s
AI Response	<3 s
Dashboard Refresh	Real-time
Interaction Latency	<100 ms
Lighthouse Score	90+

Tech Stack

Frontend

- Next.js
- React
- Tailwind CSS
- Framer Motion
- GSAP
- React Three Fiber

Backend

- FastAPI
- Firebase
- Firestore
- WebSockets

AI

- Gemini (Primary)

- Groq (Automatic Fallback)
- Vertex AI
- RAG Engine

Visualization

- Three.js
 - Google Maps Platform
 - Recharts
 - Zustand
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Acceptance Criteria

- AI Operational Summary loads automatically.
 - Digital Twin renders interactively.
 - Live KPIs update without page refresh.
 - AI Copilot supports streaming responses.
 - Gemini → Groq fallback is seamless.
 - Command Palette works globally.
 - Incident Timeline updates in real time.
 - Accessibility meets WCAG 2.2 AA.
 - Dashboard remains usable on desktop, tablet, and mobile.
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Future Enhancements

- Multi-stadium command center
- Cross-venue coordination
- Live CCTV overlays
- AI agent collaboration

- Predictive staffing optimization
 - Apple Vision Pro immersive workspace
 - Real-time drone telemetry
 - Multi-user operational collaboration
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AI Command Center Philosophy

"Operators shouldn't search for information. Information should find the operators."

ArenaMind AI transforms stadium operations from reactive monitoring into proactive, AI-driven decision intelligence, making the Command Center the living brain of FIFA World Cup stadium operations.

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